

KALAMAZOO BATTLE CREEK, MI, USA

WMO: 726357

Lat: 42.235N

Lon: 85.552W

Elev: 265

StdP: 98.19

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 94815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.0	(c) -13.9	(d) -20.9	(e) 0.6	(f) -15.7	(g) -18.0	(h) 0.8	(i) -12.7	(j) 10.8	(k) -3.0	(l) 9.8	(m) -4.0	(n) 2.9	(o) 220	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.3	22.9	30.9	22.1	29.1	21.2	24.5	29.7	23.6	28.5	22.7	27.4	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	27.6	22.1	17.3	26.8	21.2	16.4	25.7	75.2	29.7	71.6	28.5	68.1	27.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.7	(b) 8.5	(c) 7.7	(e) -20.7	(f) 34.5	(g) 4.3	(h) 1.6	(i) -23.7	(j) 35.7	(k) -26.2	(l) 36.6	(m) -28.6	(n) 37.5	(o) -31.7	(p) 38.7		
(4) 9.7	(5) 8.5	(6) 7.7	(7) -21.0	(8) 26.0	(9) 4.2	(10) 1.1	(11) -24.0	(12) 26.8	(13) -26.4	(14) 27.4	(15) -28.7	(16) 28.0	(17) -31.7	(18) 28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-4.0	-2.7	2.6	9.5	15.6	20.6	22.7	21.7	18.0	11.2	4.9	-0.9		
	DBStd	10.61	6.04	5.81	6.26	5.12	4.78	3.77	3.15	2.87	4.25	4.80	5.00	5.51		
	HDD10.0	1665	434	357	246	69	8	0	0	0	1	43	167	341		
	HDD18.3	3465	692	589	490	268	112	19	3	6	58	228	404	597		
	CDD10.0	1663	1	1	14	54	183	319	394	362	239	79	13	2		
	CDD18.3	421	0	0	1	3	28	88	139	109	47	6	0	0		
	CDH23.3	3662	0	0	9	32	272	771	1281	857	396	44	0	0		
	CDH26.7	1113	0	0	1	4	71	248	444	227	111	7	0	0		
(14) Wind	WSAvg	3.5	4.1	4.0	4.0	4.0	3.3	3.0	2.8	2.5	2.7	3.3	3.8	4.0		
(15) Precipitation	PrecAvg	924	52	40	61	87	84	95	97	90	98	74	73	69		
	PrecMax	1191	118	96	101	135	161	211	202	285	269	179	164	130		
	PrecMin	683	14	1	4	20	15	18	19	14	0	22	10	15		
	PrecStd	114	31	27	28	31	38	53	46	58	60	33	37	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	16.2	24.3	27.0	31.2	33.5	34.9	32.7	32.2	27.7	20.5	15.9		
		MCWB	11.4	11.5	17.2	16.8	20.9	23.2	24.2	23.2	22.0	20.7	14.3	13.9		
	2%	DB	9.6	11.4	19.6	23.6	28.6	31.4	32.4	30.8	29.8	24.1	17.3	12.4		
		MCWB	7.5	8.4	14.2	15.3	19.5	21.9	23.4	22.8	21.7	17.4	12.7	10.7		
	5%	DB	6.3	7.6	16.0	21.2	26.5	29.5	30.9	29.0	27.5	21.7	15.0	8.9		
		MCWB	4.5	5.0	11.5	13.7	18.5	20.9	22.4	21.5	19.9	16.5	11.2	7.0		
	10%	DB	3.9	5.0	12.4	18.5	24.0	27.7	29.0	27.6	25.6	19.1	12.6	6.3		
		MCWB	2.4	3.0	8.6	12.1	17.5	20.3	21.5	20.9	18.9	15.0	9.4	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.8	17.8	19.0	22.6	24.7	26.0	25.2	24.0	21.7	15.6	14.1		
		MCDB	12.8	14.6	24.2	24.3	28.5	30.8	32.4	30.3	29.7	26.7	18.4	15.9		
	2%	WB	7.9	8.7	15.0	16.7	21.0	23.5	24.8	24.0	22.6	19.4	13.8	10.7		
		MCDB	9.1	11.0	18.6	21.4	26.4	29.0	30.2	28.4	27.3	22.2	16.5	12.1		
	5%	WB	4.6	5.4	12.2	14.9	19.8	22.4	23.9	23.0	21.4	17.3	11.8	7.3		
		MCDB	6.2	7.4	15.3	19.7	24.6	27.7	28.9	27.3	25.5	20.3	14.5	8.5		
	10%	WB	2.5	3.0	8.7	13.0	18.5	21.5	22.9	22.1	20.1	15.4	9.7	4.5		
		MCDB	3.9	4.9	11.8	17.9	22.9	26.2	27.5	26.1	24.1	18.8	12.6	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.8	9.5	11.3	11.1	10.8	11.1	10.7	11.3	9.9	8.0	6.5		
		MCDBR	8.8	10.7	13.7	14.4	13.4	12.4	12.3	11.9	12.9	13.0	11.3	9.4		
		MCWBR	7.6	8.4	9.0	8.2	6.8	5.4	5.2	5.3	6.0	7.5	8.1	8.0		
		MCDBR	8.2	10.2	12.2	12.5	11.4	10.4	10.4	9.6	10.6	11.0	10.1	8.7		
	5% WB	MCWBR	7.5	8.6	8.8	8.4	6.7	5.4	5.3	5.1	5.7	7.8	8.1	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.353	0.350	0.389	0.421	0.439	0.431	0.437	0.394	0.354	0.319	0.302		
	taud		2.366	2.177	2.339	2.254	2.210	2.201	2.229	2.209	2.307	2.419	2.467	2.401		
	Ebn at Noon		845	832	893	878	855	838	841	822	834	831	816	804		
	Edh at Noon		81	116	112	131	141	142	137	136	114	91	74	73		
(44) All-Sky Solar Radiation	RadAvg		1.36	2.20	3.50	4.64	5.37	6.08	6.17	5.32	4.29	2.73	1.68	1.13		
	RadStd		0.13	0.32	0.35	0.43	0.53	0.40	0.28	0.29	0.39	0.29	0.24	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page